How much does wind cost? A paired calm/windy synthetic RGB-D benchmark that separates occlusion, sensor and wind error in simulated 3D foliage reconstruction

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Abstract

Neural scene representations such as 3D Gaussian Splatting and neural radiance fields are increasingly used to reconstruct plants for phenotyping, but they assume a static scene, while foliage is never still outdoors or near fans and ventilation. Existing plant datasets cannot tell how much of the reconstruction error is caused by the method and occlusion, by the depth sensor, or by leaf motion, because ground truth for moving leaves is unavailable. We present SynthTree, a fully scripted generator of paired calm/windy RGB-D sequences of potted saplings, modelled on a common hand-held RGB-D capture protocol (Intel RealSense D435i, fan-generated wind). Every tree is rendered at several canopy wind speeds with a bit-identical camera trajectory; leaves are real geometry with unique identities, and a closed-form wind model (hierarchical branch sway and rigid leaf flutter) makes the exported per-leaf rigid transforms exact by construction. Automatic checks on all 40 sequences confirm sub-pixel consistency of poses, intrinsics, depth and per-leaf motion (re-projection error ≤ 0.0036 px; per-leaf transforms predict the rendered optical flow to 0.007 px). We further propose a three-way error decomposition with an explicit interaction term and a geometry-first evaluation protocol for Gaussian reconstructions. In 108 training runs on four trees (one per leaf type), wind at 5 m/s reduces the leaf F-score of 3D Gaussian Splatting (Nerfstudio splatfacto) by 0.341 and of a depth-supervised NeRF by 0.348, and raises the per-leaf centroid error of splatfacto by 12.0 mm, while image-space PSNR changes by only -7 %: in this setting, view-synthesis metrics hide most of the geometric damage. Simulated sensor

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noise, in contrast, mainly trades precision for coverage. Using the exported per-leaf motion, we further show that, for our static 3DGS baseline, evaluating against the rest pose over-states the wind error by 34 %, and that a robust trimmed loss helps in all conditions but does not reduce the wind component. A deformation-field Gaussian baseline collapses to a uniform shift; a follow-up at the full 30 fps shows that this is a failure of the motion model rather than of temporal sampling. Per-part rigid motion removes the collapse and reduces the wind part of the chamfer error by 19 %, but even with ground-truth leaf grouping about three quarters of it remain, and four times denser sampling brings no geometric gain. All experiments are synthetic, with sensor and wind models set to plausible, literature-informed defaults and without motion blur or rolling shutter; the findings hold in simulation under these models and remain to be confirmed on real captures.

Keywords: 3D plant reconstruction, Gaussian splatting, neural radiance fields, wind, RGB-D sensor noise, synthetic data, phenotyping

1. Introduction

Three-dimensional reconstruction of plants underpins non-destructive phenotyping of leaf area, leaf angle distribution and canopy architecture (Paulus, 2019). Recent neural representations, neural radiance fields (Mildenhall et al., 2020; Müller et al., 2022) and 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023), produce photorealistic novel views from ordinary image or RGB-D sequences and are rapidly being adopted for plants. All of them assume a static scene. Leaves, however, flutter at a few hertz in a light breeze and whole branches sway (de Langre, 2008), and capturing a full orbit around a sapling takes tens of seconds.

When a reconstruction of a real plant is poor, the cause is ambiguous: thin, self-occluding foliage is intrinsically hard for the method; consumer depth sensors add noise, holes and flying pixels at leaf boundaries (Khoshelham and Elberink, 2012; Keselman et al., 2017); and leaf motion violates the static-scene assumption. These effects cannot be separated on real data, because the true geometry of a moving canopy is unknown, and existing benchmarks either capture static scenes (Jensen et al., 2014; Knapitsch et al., 2017; Sturm et al., 2012) or provide plant point clouds without per-leaf motion (Schunck et al., 2021).

We address this with a controlled synthetic benchmark, modelled on a common capture protocol (a hand-held orbit around a potted sapling with

a consumer RGB-D camera and a fan), in which every factor is controlled and every leaf is tracked exactly. Our contributions are:

1. **SynthTree**, an open, fully scripted pipeline (procedural saplings, exact procedural wind, D435i-like RGB-D rendering, sensor-noise simulation with a calibration procedure verified on synthetic recordings) that produces *paired* calm/windy sequences: same tree, bit-identical camera path, only the wind speed differs.
2. **Exact per-leaf ground truth under wind**: leaf identity maps, per-leaf rigid transforms, optical flow and canonical correspondences, validated to sub-pixel accuracy on all sequences.
3. **A three-way error decomposition** into occlusion/method, sensor and wind components with an explicit interaction term, and a geometry-first evaluation protocol for Gaussian reconstructions that is robust to floaters and floor splats.
4. **A benchmark** showing that, in simulation under default sensor and wind models and for the methods tested, wind dominates the sensor effect, view-synthesis metrics substantially under-report the geometric degradation of foliage, and rest-pose evaluation over-states the wind error of a static reconstruction; time-resolved per-leaf ground truth corrects this and allows dynamic methods to be scored.
5. **A sampling-versus-modelling analysis** of dynamic Gaussian reconstruction under wind: the collapse of a deformation field persists at 30 fps, per-part rigid motion removes it but recovers little geometry even with ground-truth leaf groups, and denser sampling does not help without motion blur.

Code and data are released (Section 7).

2. Related work

3D plant reconstruction and phenotyping. Laser scanning, multi-view stereo and RGB-D cameras are established for measuring plant geometry (Paulus, 2019); Pheno4D (Schunck et al., 2021) provides spatio-temporal point clouds of growing plants with organ labels. Spatio-temporal analysis of plants has so far addressed growth over hours to days, tracking and matching organs between successive scans (Li et al., 2013); at that time scale wind-induced motion within one scan is treated as noise, not measured. Wind-induced motion of plants is well characterised biomechanically, from slow branch sway to leaf flutter at several hertz (de Langre, 2008), but its effect on reconstruction accuracy has not been quantified. Existing datasets are acquired in still air and, being real, cannot provide the true geometry of moving leaves.

Neural reconstruction of static and dynamic scenes. NeRF (Mildenhall et al., 2020) and its accelerated variants (Müller et al., 2022), and 3DGS (Kerbl et al., 2023) and its surface-oriented variant 2DGS (Huang et al., 2024a), reconstruct static scenes; Nerfstudio (Tancik et al., 2023) provides the implementations we benchmark. Dynamic extensions model deformation over time (Pumarola et al., 2021; Luiten et al., 2024; Wu et al., 2024; Yang et al., 2024). Per-part and motion-basis variants assign Gaussians to superpoints with rigid motion (Wan et al., 2024), to sparse control points (Huang et al., 2024b) or to a small set of shared SE(3) motion bases (Wang et al., 2024), or give each Gaussian a polynomial and Fourier trajectory (Lin et al., 2024); monocular dynamic reconstruction is further limited by how far the camera moves relative to the scene motion (Gao et al., 2022). Gaussian-splatting SLAM (Matsuki et al., 2024) tracks RGB-D cameras. Robust variants instead treat inconsistent content as outliers, either with per-image appearance and transient embeddings (Martin-Brualla et al., 2021) or by down-weighting pixels with large residuals (Sabour et al., 2023); our trimmed-loss mitigation follows the latter idea. How these methods degrade on fluttering foliage has not been quantified with ground truth.

Synthetic data and benchmarks. Rendering pipelines such as BlenderProc (Denninger et al., 2023) and ICL-NUIM (Handa et al., 2014) provide exact ground truth for rigid indoor scenes, and procedural plant models have a long history (Prusinkiewicz and Lindenmayer, 1990; Weber and Penn, 1995; Runions et al., 2007). To our knowledge no benchmark provides paired static/dynamic foliage with per-leaf motion ground truth and a consumer RGB-D sensor model.

Depth-sensor noise. Depth noise of structured-light and active-stereo cameras grows approximately quadratically with distance and concentrates at depth discontinuities (Khoshelham and Elberink, 2012; Nguyen et al., 2012; Keselman et al., 2017), which is precisely where thin leaves are.

3. The SynthTree generator

Figure 1 shows the 10 saplings of the released dataset and Fig. 3 the per-frame products. The pipeline is pure Python/NumPy except for rendering (Blender, headless); identical configuration and seed give bit-identical geometry, poses and ground truth, which we verified across two Blender LTS versions with different Python/NumPy versions.

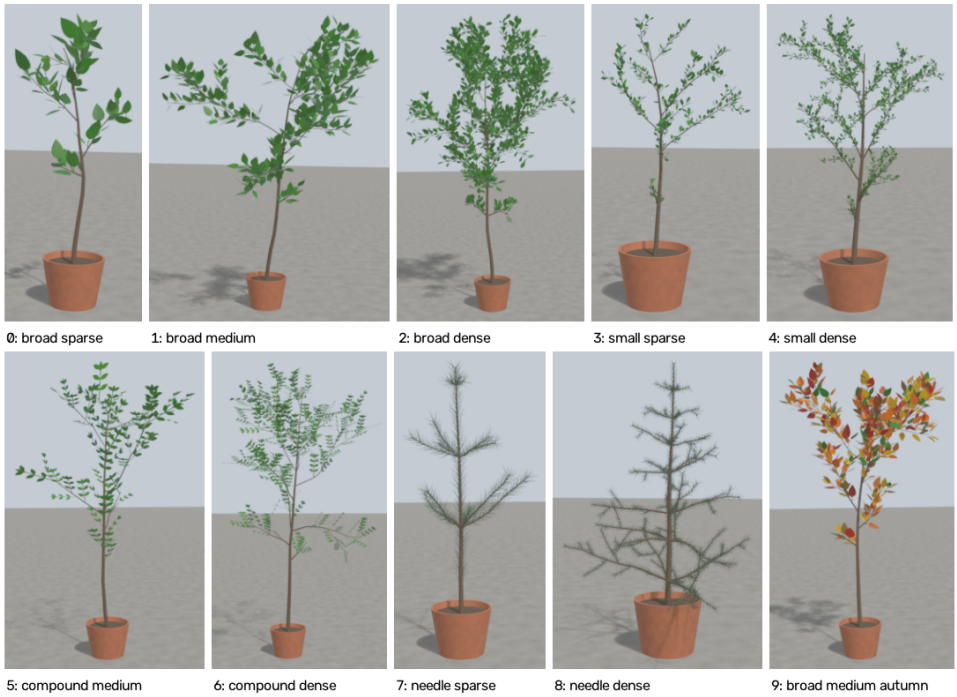


Figure 1: The 10 procedural saplings (calm, first frame): broad, small, pinnate-compound and needle leaves at sparse to dense canopy densities, including an autumn-coloured variant for vegetation-mask robustness. Leaves are real meshes, not alpha-textured cards.

3.1. Procedural saplings with leaf identities

Each tree is a skeleton of nodes (position, radius, parent) grouped into branches of increasing order, grown with stochastic wander, phototropic upturn and phyllotactic or whorled branching (Weber and Penn, 1995). Wood is meshed as parallel-transported tubes. Leaf sites are spaced along branch segments; a leaf is attached at the bark surface of a segment and inherits that segment's frame. Leaf blades are real triangulated surfaces with species-dependent outlines (ovate, elliptic, obovate, lanceolate), cupping and droop; compound leaves consist of a rachis and leaflets, and needles are closed triangular prisms. Every leaf has an integer identity, an attachment point, a parent branch, a rest frame, an area and a centroid, stored with the canonical (rest) mesh (Table 1). Heights range from 0.82 to 1.98 m and leaf counts from 62 to 8,192.

Table 1: Dataset composition. Height includes the pot; triangles count wood and leaves. The experiments use trees 1, 3, 5 and 7 (one per leaf type); trees 8 and 9 are designated as a test split for future generalisation studies and are not used in this paper.

ID	Preset	Leaf type	Height (m)	Leaves	Leaf area (m^2)	Triangles	Split
0	broad_sparse	broad	1.00	86	0.161	2,890	train
1	broad_medium	broad	1.54	624	0.874	17,234	train
2	broad_dense	broad	1.98	2,591	2.460	65,376	train
3	small_sparse	small	0.87	535	0.064	9,190	train
4	small_dense	small	0.82	1,865	0.119	29,992	train
5	compound_medium	compound	1.44	62	0.257	6,378	train
6	compound_dense	compound	1.73	217	0.505	27,044	train
7	needle_sparse	needle	1.14	1,672	0.078	28,107	train
8	needle_dense	needle	1.11	8,192	0.146	136,748	test
9	broad_medium_autumn	broad	1.47	557	0.829	15,366	test

3.2. Exact procedural wind

We deliberately do not use physics simulation or a node-based deformer: the same closed-form function both moves the rendered vertices and produces the exported ground truth, so the two are identical by construction.

Branch sway. Every skeleton node j receives a bend rotation

$$\Delta_j(t) = \mathrm{Rot}\big(\mathbf{b}, \theta_j(t)\big)\,\mathrm{Rot}\big(\mathbf{w}, \psi_j(t)\big), \tag{1}$$

where $\mathbf{w}$ is the horizontal wind direction and $\mathbf{b} = \mathbf{e}_z \times \mathbf{w}$. Rotations are accumulated from the root by forward kinematics, $A_j = A_{p(j)}\Delta_j$ and $\mathbf{p}_j = \mathbf{p}_{p(j)} + A_{p(j)}(\mathbf{p}^0_j - \mathbf{p}^0_{p(j)})$, so displacement grows with height and branching order. The bend and cross-wind angles are

$$\theta_j = c_j\, g_b(U_e)\,[\alpha_s + \alpha_o \sin(2\pi f_B t + \phi_B)], \tag{2}$$

$$\psi_j = c_j\, g_b(U_e)\,\alpha_c \sin(2\pi r f_B t + \phi'_B), \tag{3}$$

with frequency f_B and phase ϕ_B shared by all nodes of a branch B (coherent sway), and a compliance $c_j \propto (r_0/r_j)\,\ell_j$ that favours thin, long segments.

Leaf flutter. Each leaf rotates rigidly about its petiole attachment,

$$\begin{aligned} R_\ell = \mathrm{Rot}(\mathbf{m}_\ell, g_l\beta_m)\,\mathrm{Rot}\big(\mathbf{x}_\ell, g_l\beta_t \sin(2\pi r_\ell f_\ell t + \phi'_\ell)\big) \\ \mathrm{Rot}\big(\mathbf{y}_\ell, g_l\beta_p \sin(2\pi f_\ell t + \phi_\ell)\big), \end{aligned} \tag{4}$$

with a per-leaf frequency $f_\ell \in [2.5, 9]\,\mathrm{Hz}$, mean down-wind reconfiguration about $\mathbf{m}_\ell = \mathbf{n}_\ell \times \mathbf{w}$, and is carried by its node: $A_\ell = A_{n(\ell)}R_\ell$. The leaf's rigid transform from the canonical pose, $\mathbf{x}_t = A_\ell \mathbf{x}_0 + \mathbf{t}_\ell$, is exported per frame.

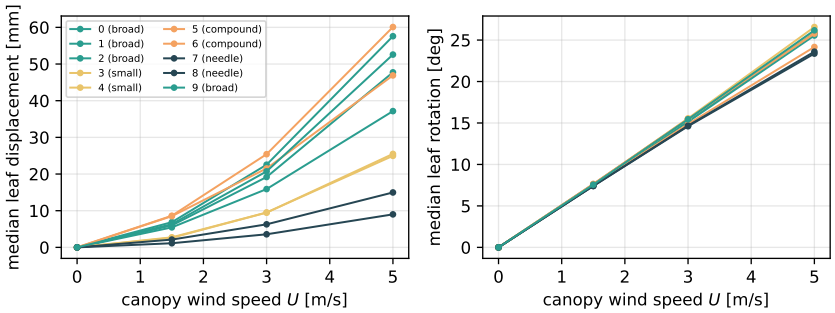


Figure 2: Ground-truth leaf motion per tree versus canopy wind speed (median over leaves and frames).

Amplitude laws and gusts. $g_b(U) = (U/U_r)^2$ follows the quadratic dependence of drag on speed for small elastic deflections; $g_l(U) = s \tanh\big((U/U_r)/s\big)$ lets flutter grow roughly linearly and saturate. Gusts modulate the speed, $U_e(t) = U \max\{0,\, 1 + \gamma\, n(t)\}$, with a smooth seeded noise $n(t)$. All angles vanish exactly at $U = 0$ and rotations are evaluated element-wise, so calm sequences are the rest pose bit-for-bit. Phases depend only on the tree seed, so different wind levels of a tree differ only in amplitude. Over 150 frames, the median leaf displacement at 5 m/s ranges from 9 to 60 mm across trees, with median leaf rotations of 23–27° (Fig. 2). The amplitudes are plausible defaults chosen so that leaves move by millimetres in light wind and by several centimetres at 5 m/s; they are configurable, and a calibration procedure against static-camera fan recordings is provided (Section 3.5). Because absolute amplitudes are not tied to a specific plant and fan, the resulting leaf motion is reported alongside U (Fig. 2).

3.3. D435i-like rendering and ground-truth passes

The camera uses D435i-like pinhole intrinsics for the 848×480 colour stream (nominal values, reproduced exactly in Blender through focal length, principal point via lens shift and pixel aspect, and replaceable by the calibrated values of a specific device) and a hand-held orbit with seeded smooth jitter; the trajectory depends only on the tree seed, so it is bit-identical across wind levels. Time is explicit: frame i is rendered at $t = i\,k/\mathrm{fps}$ for both wind and camera.

RGB is rendered with EEVEE or Cycles and converted to 8-bit sRGB with the exact transfer function. Ground truth is rendered in a separate Cycles pass with one sample per pixel and a 0.01-pixel box filter, so every

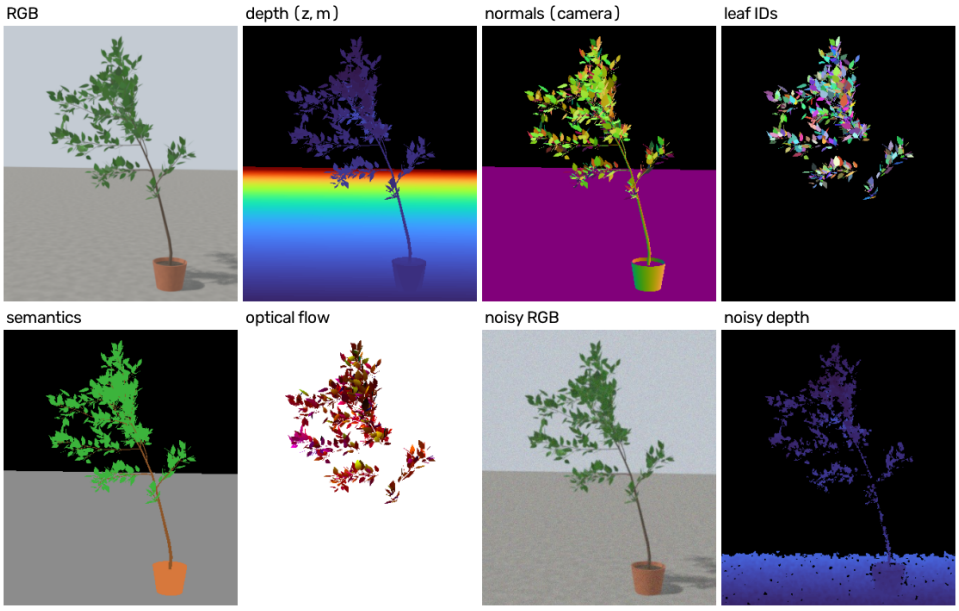


Figure 3: Per-frame products for a broad-leaved sapling at $U = 5\,\mathrm{m/s}$. Clean products are exact; noisy RGB and depth follow the sensor model of Section 3.4.

value is the first hit of the ray through the pixel centre and is never blended across edges. Shader outputs store per-pixel world position, normal, leaf identity, semantic class, rest position and next-frame position. Depth is computed as $z = (\mathbf{P} - \mathbf{C}) \cdot \mathbf{o}$ from the world position, so it is z-depth by construction. Exact optical flow is obtained by projecting the interpolated next-frame position, which also covers non-rigid wood deformation. Motion blur and rolling shutter can be rendered by sub-frame and row-band integration, but *both are disabled in every experiment reported here*: all RGB frames are instantaneous global-shutter images, which is optimistic relative to real hand-held capture of moving foliage (Section 6).

3.4. Sensor noise model

Clean data are always kept and a noisy copy is derived. Depth undergoes resampling to the sensor's effective depth resolution; spatially correlated noise with $\sigma(z) = a + bz^2$ or, alternatively, Gaussian disparity noise with sub-pixel quantisation; flying pixels that interpolate foreground and background depth at discontinuities; holes whose probability increases near edges, on structures thinner than a few pixels and at grazing angles; range limits; misregistration; and quantisation to millimetres. RGB receives an

Table 2: Automatic validation over all released sequences.

Check (all 40 sequences)	Statistic	Value
World position re-projects onto its own pixel centre	max error	0.0036 px
Cross-frame re-projection of exported depth (static geometry)	worst median	0.002 px
Back-projected floor lies on $z = 0$ (image corners)	worst median	3 μm
same test if depth were ray distance	median	147 mm
16-bit PNG depth vs. float EXR depth	max difference	0.50 mm
Leaf-ID pixels not in table / inconsistent with semantics	total	0 px
Per-leaf transforms predict rendered flow (leaf pixels)	worst median	0.007 px
Rendered flow of static geometry	worst median	0.006 px

auto-exposure random walk, slight blur, shot and read noise and JPEG compression. Noise draws are keyed to the tree and frame, so they are paired across wind levels.

3.5. Calibration tooling

All experiments in this paper use D435-like default noise parameters ($\sigma(z) = 0.5\,\mathrm{mm} + 3.8\,\mathrm{mm\,m^{-2}}z^2$, i.e. about 4 mm at 1 m) and default wind amplitudes. To adapt SynthTree to a specific device and fan, the generator includes calibration procedures. The noise parameters are fitted from recordings of a flat textured wall at six distances (plane-fit residuals give a, b and the spatial correlation; invalid pixels give the random hole rate) and of leaves against a distant background (vegetation silhouettes give flying-pixel, edge-hole and thin-structure-hole rates by fixed-point matching of simulated statistics). On synthetic recordings with known parameters, the procedure recovers a, b and the correlation length within 3 % and the flying-pixel and hole rates within 2 %. Wind amplitudes can be calibrated analogously by matching the median metric leaf speed (optical flow scaled by depth) of static-camera fan clips, iterating until all wind levels agree within 3 %; a self-test with a known amplitude factor of 1.6 recovers 1.57.

3.6. Validation

Every sequence is validated automatically on the exported products (Table 2); all 40 of 40 sequences pass. The decisive check for depth convention is geometric: back-projecting the depth of the floor yields points within 3 μm of the plane $z = 0$ even in the image corners, whereas interpreting the same values as ray distance would displace them by 147 mm.

4. Benchmark protocol

Data and splits. Each tree is rendered at four canopy wind speeds, $U \in \{0, 1.5, 3, 5\}$ m/s, where $U = 0$ is the calm sequence, giving 40 sequences for 10 trees; each sequence is a single orbit of 150 frames at the sensor resolution (848×480). Every eighth frame is held out for view-synthesis evaluation. All experiments use trees 1, 3, 5 and 7, one per leaf type (Table 1); in tables, low, medium and high denote $U = 1.5$, 3 and 5 m/s. No generalisation experiments are reported. Motion blur and rolling shutter are disabled throughout.

Methods. We evaluate splatfacto (3DGS) and depth-nerfacto (NeRF with depth supervision) from Nerfstudio (Tancik et al., 2023), trained with the ground-truth poses in metric world coordinates (no re-centring or re-scaling) and initialised from back-projected sensor depth of the same variant (clean or noisy). To test whether modelling motion helps, we additionally implement, on the same gsplat rasteriser, a static 3DGS and a deformable 3DGS that share initialisation, losses (0.8 L1 + 0.2 D-SSIM), densification, background and iteration budget (10,000); the deformable variant adds a time-conditioned MLP (8 layers, 256 units) that predicts per-Gaussian offsets of position, rotation and scale from positionally encoded canonical position and time (Yang et al., 2024), enabled after 2,000 static warm-up iterations. As a simple mitigation that requires no motion model, a third variant replaces the L1 term by a trimmed L1 that ignores the 20 % of pixels with the largest residual in each training view (after 1,000 iterations), so that moving leaves the static model cannot explain do not pull Gaussians into blurred compromises.

Geometry-first evaluation. Reconstructions are compared with the canonical leaf surface. Each reconstructed point is attributed to its nearest ground-truth surface (leaf, wood, pot/soil or floor); points farther than 3 cm from every surface are floaters and are reported separately. For Gaussians, precision-side quantities (accuracy, precision, floaters, per-leaf centroids, leaf angles) use Gaussian centres, so each splat counts once, whereas coverage-side quantities (completeness, recall, recovered leaf area, leaf coverage) use samples spread over each Gaussian's footprint, because centres alone under-cover thin surfaces. We report the F-score at $\tau = 5$ mm, a capped chamfer distance, the relative leaf-area error, the 1-D Wasserstein distance between leaf inclination distributions (Ross, 1981), and the median per-leaf centroid error. Needles are thinner than a pixel, so for conifers per-leaf quantities are

computed per shoot (all needles of a twig). View-synthesis PSNR is reported on held-out views.

Rest-pose and posed evaluation. The canonical (rest) surface is the natural target when the goal is the plant's architecture, but under wind the true surface at any instant is displaced from it. Because SynthTree exports the rigid transform of every leaf at every frame, we also evaluate *posed*: the leaf surface is posed with the transforms of a held-out frame and compared with the reconstruction at that frame's time (the same reconstruction for static methods, the deformed one for dynamic methods), averaged over four held-out frames.

Three-way decomposition. For a metric E oriented so that larger is worse,

$$E_{\text{method}} = E(\text{calm}, \text{clean}), \tag{5}$$

$$E_{\text{sensor}} = E(\text{calm}, \text{noisy}) - E(\text{calm}, \text{clean}), \tag{6}$$

$$E_{\text{wind}}(U) = E(U, \text{noisy}) - E(\text{calm}, \text{noisy}), \tag{7}$$

$$E_{\text{int}}(U) = \big[E(U, \text{noisy}) - E(U, \text{clean})\big] - E_{\text{sensor}}. \tag{8}$$

The interaction term measures how far sensor and wind errors are from additive. Values are averaged over seeds per configuration, and mean and standard deviation over trees are reported.

5. Results

We report 108 training runs in total (Table 3): 60 in the benchmark and 48 in the baseline study of Section 5. The benchmark trains splatfacto on 4 trees (one per leaf type: broad, small, compound, needle) under six conditions (noisy sensor at all four wind speeds, clean sensor at calm and 5 m/s) with 2 seeds each, and depth-nerfacto on a broad-leaved and a needle tree under the same six conditions with one seed. The mean standard deviation of the leaf F-score between seeds is 0.014, an order of magnitude below the wind effects reported below. Depth-nerfacto is restricted to two trees and one seed because each run takes 82 min on the available hardware; its conclusions are therefore indicative.

Calm and clean (occlusion/method error). Even with exact poses and noise-free depth, splatfacto reaches only F@5mm = 0.379 and a chamfer distance of 19.3 mm on leaves (Table 5), and inclines leaves more steeply than they are; this is the intrinsic difficulty of thin, self-occluding foliage. Depth-supervised depth-nerfacto recovers markedly better calm geometry (F@5mm = 0.558, chamfer distance 6.4 mm) with almost no floaters.

Table 3: Accounting of all training runs reported in this paper. A superseded first round of 16 deformable runs with a six-band temporal encoding (Section 5) is not included.

Study	Method	Trees	Wind U [m/s] per sensor variant	Seeds	Runs
Baseline study	deformable 3DGS (10-band)	1, 3, 5, 7	noisy: calm, 1.5, 3, 5	1	16
Baseline study	static 3DGS + trimmed loss	1, 3, 5, 7	noisy: calm, 1.5, 3, 5	1	16
Baseline study	static 3DGS (ours, gsplat)	1, 3, 5, 7	noisy: calm, 1.5, 3, 5	1	16
Benchmark	depth-nerfacto (Nerfstudio)	1, 7	noisy: calm, 1.5, 3, 5; clean: calm, 5	1	12
Benchmark	splatfacto (Nerfstudio)	1, 3, 5, 7	noisy: calm, 1.5, 3, 5; clean: calm, 5	2	48
Total reported training runs					108

Table 4: Benchmark results. splatfacto: mean over 4 trees and 2 seeds; depth-nerfacto: mean over 2 trees, 1 seed. Low, medium, high: $U = 1.5$, 3, 5 m/s. F@5mm, Prec., Rec.: leaf F-score, precision and recall at $\tau = 5$ mm; CD: capped chamfer distance; Area err: relative leaf-area error; Angle W1: leaf inclination distribution distance; Floaters: fraction of reconstructed points farther than 3 cm from every surface.

Method	Condition	F@5mm↑	Prec.↑	Rec.↑	CD [mm]↓	Centroid [mm]↓	Area err↓	Angle W1 [deg]↓	Floaters↓	PSNR [dB]↑
depth-nerfacto	calm / clean	0.558	0.75	0.47	6.4	5.6	0.53	10.1	0.01	35.5
	calm / noisy	0.525	0.66	0.45	6.7	6.5	0.55	10.9	0.01	26.2
	low / noisy	0.424	0.59	0.35	7.6	8.4	0.65	12.1	0.01	26.9
	medium / noisy	0.151	0.16	0.18	19.8	14.1	0.82	15.6	0.10	25.6
	high / noisy	0.178	0.22	0.15	17.0	18.1	0.85	12.5	0.04	26.2
	high / clean	0.193	0.30	0.15	14.6	14.5	0.85	12.3	0.04	29.2
splatfacto	calm / clean	0.379	0.38	0.42	19.3	6.7	0.58	12.6	0.19	36.1
	calm / noisy	0.439	0.39	0.60	17.8	8.6	0.40	12.0	0.32	29.1
	low / noisy	0.306	0.23	0.52	19.3	10.8	0.48	15.4	0.34	28.7
	medium / noisy	0.168	0.12	0.31	22.2	13.8	0.69	16.7	0.35	27.9
	high / noisy	0.099	0.07	0.19	25.9	20.6	0.81	17.7	0.43	27.1
	high / clean	0.098	0.08	0.14	26.0	19.4	0.86	16.7	0.19	29.5

Wind. Wind is the dominant degradation. For splatfacto at 5 m/s the leaf F-score drops by 0.341, the chamfer distance increases by 8.1 mm and the per-leaf (per-shoot for the conifer) centroid error by 12.0 mm (Fig. 4); the seed-averaged F-score decreases monotonically with wind speed on all four trees, and per seed in 7 of 8 tree–seed pairs. Depth-nerfacto loses a comparable amount (F-score −0.348, chamfer distance +10.3 mm), so its better calm geometry does not protect it against leaf motion; for the conifer its single-seed curve is not monotonic, which more seeds must resolve. Broad-, small- and compound-leaved trees degrade most; the needle tree, whose shoots move less (Fig. 2), degrades least. Per tree, the median leaf displacement at 5 m/s, the absolute F-score drop and the relative drop of splatfacto are: tree 1: 48 mm, 0.48 (82 %); tree 3: 25 mm, 0.31 (85 %); tree 5: 47 mm, 0.44 (91 %); tree 7: 15 mm, 0.14 (42 %). The absolute drop follows the amount of leaf motion (Spearman $\rho = 1.00$; $\rho = 0.40$ for the relative drop), but with $n = 4$ trees this is descriptive, not a significance claim; leaf type and canopy density vary together with motion across our trees.

Sensor. For splatfacto the sensor term is negative in F-score (-0.060) and chamfer distance (-1.5 mm), i.e. noisy input appears to *help*. The compo-

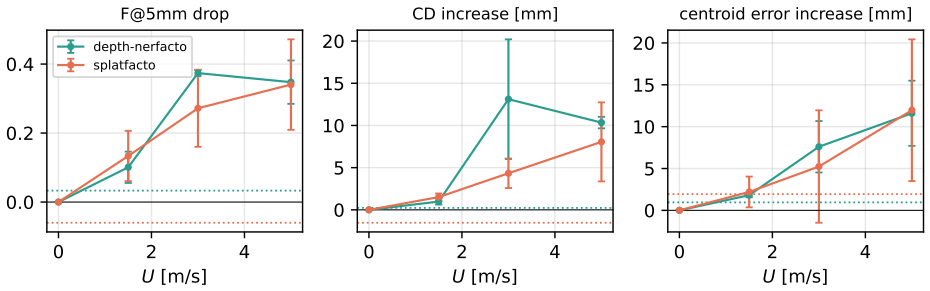


Figure 4: Wind error versus canopy wind speed. Points: mean over trees after averaging seeds; error bars: ± one standard deviation over trees (4 for splatfacto, 2 for depth-nerfacto), dominated by tree-to-tree differences in leaf type and motion (between-seed std of F ≈ 0.014). Dotted lines mark the sensor error.

nents explain why: with noisy RGB-D input, splatfacto converges to 2.0× fewer Gaussians near the tree (14,873 vs. 30,249), which are larger, so recall rises from 0.42 to 0.60 while precision is unchanged (0.38 vs. 0.39) and the floater fraction grows from 0.19 to 0.32. Sensor noise thus trades geometric precision for coverage, and aggregate scores such as the F-score or a capped chamfer distance can mistake this for an improvement; we therefore recommend reporting precision and recall separately. For depth-nerfacto, which is supervised by the depth itself, the sensor term is positive (0.033 in F-score) as expected. Because the noise parameters are D435-like defaults rather than values calibrated to one device, the magnitude of the sensor term is specific to this noise model; the qualitative precision–coverage trade-off is the transferable finding.

Interaction. For splatfacto at 5 m/s the interaction is $+1.4 \pm 1.5$ mm in chamfer distance and -0.8 ± 2.1 mm in centroid error, against wind terms of +8.1 and +12.0 mm (Table 5); for depth-nerfacto (two trees) it is $+2.2 \pm 3.1$ mm in chamfer distance. The interaction is thus small relative to the wind term, but its tree-to-tree spread is as large as its mean: within the resolution of four trees we see no evidence of strong non-additivity, which is weaker than establishing that sensor and wind errors are additive.

View synthesis hides geometric damage. From calm to 5 m/s, PSNR on held-out views changes by -7 % while the leaf F-score changes by -75 % (Fig. 5). Methods that average moving leaves into blurred but plausible appearance are rewarded by image metrics; per-leaf geometric ground truth is needed to expose the error.

Table 5: Three-way error decomposition (splatfacto: mean ± std over 4 trees after averaging 2 seeds; depth-nerfacto: 2 trees, 1 seed). †The occlusion/method row gives absolute values in the calm, clean condition (F@5mm, CD, centroid error). All other rows are signed error increases as defined in Section 4: positive means worse, so a positive F entry is a *decrease* of the F-score, and a negative entry (e.g. the splatfacto sensor term) is an apparent improvement. Large standard deviations reflect tree-to-tree variability (leaf type, density), not seed noise.

Method	Component	F@5mm drop	CD increase [mm]	Centroid increase [mm]
depth-nerfacto	Occlusion/method†	0.558 ± 0.078	6.4 ± 0.6	5.6 ± 1.5
	Sensor	+0.033 ± 0.090	+0.2 ± 1.1	+1.0 ± 0.3
	Wind, $U = 1.5$ m/s	+0.101 ± 0.045	+1.0 ± 0.4	+1.8 ± 0.2
	Wind, $U = 3$ m/s	+0.374 ± 0.009	+13.1 ± 7.1	+7.6 ± 3.1
	Wind, $U = 5$ m/s	+0.348 ± 0.063	+10.3 ± 0.7	+11.6 ± 3.9
	Interaction, $U = 5$	-0.018 ± 0.089	+2.2 ± 3.1	+2.6 ± 4.5
splatfacto	Occlusion/method†	0.379 ± 0.117	19.3 ± 4.1	6.7 ± 5.7
	Sensor	-0.060 ± 0.055	-1.5 ± 1.5	+1.9 ± 3.1
	Wind, $U = 1.5$ m/s	+0.134 ± 0.073	+1.5 ± 0.4	+2.2 ± 1.8
	Wind, $U = 3$ m/s	+0.272 ± 0.112	+4.3 ± 1.8	+5.2 ± 6.7
	Wind, $U = 5$ m/s	+0.341 ± 0.131	+8.1 ± 4.7	+12.0 ± 8.5
	Interaction, $U = 5$	+0.059 ± 0.054	+1.4 ± 1.5	-0.8 ± 2.1

Rest pose versus posed ground truth. For static 3DGS (4 trees, noisy sensor), the calm-to-5 m/s drop in F-score is 0.355 against the rest pose but only 0.265 against the posed leaves (Table 6 and Fig. 6): at 5 m/s the reconstruction matches the moving canopy (F@5mm = 0.191) better than the rest pose (0.108). Posed F exceeds rest-pose F in 10 of 12 windy tree–wind conditions. A static reconstruction of a windy plant converges towards its mean, down-wind-deflected shape, so rest-pose evaluation over-states the wind error by 34 % here; the time-resolved per-leaf ground truth is needed to separate deflection from blur. This figure is specific to our static 3DGS on four trees with one seed and noisy input; whether other static methods behave alike was not tested.

Deformable baseline. The deformable model does not recover the geometry lost to wind: at 5 m/s its posed F@5mm is 0.187 against 0.191 for the static model, and held-out PSNR rises by only 0.1–0.3 dB (Table 6 and Fig. 6). The learned deformation explains why. Its per-Gaussian offsets have a median of 0.9–4.0 mm with a 95th-percentile-to-median ratio of 1.00, and the offset is no larger in strong wind (2.4 mm) than in calm air (2.8 mm): the field collapses to a nearly uniform shift instead of leaf-specific, wind-dependent

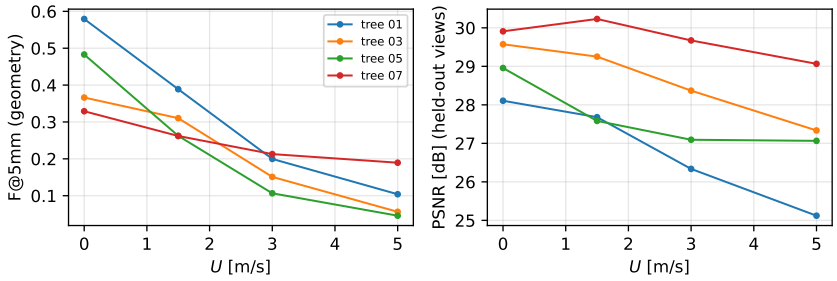


Figure 5: Geometry (left) versus view-synthesis quality (right) under increasing wind, splatfacto, noisy sensor; mean ± std over 4 trees after averaging seeds.

motion. We used a temporal encoding with ten frequency bands, because the six bands of Yang et al. (2024) limit temporal variation to about 0.8 Hz over a 20 s orbit, below branch-sway (0.6–2 Hz) and flutter (2.5–9 Hz) frequencies; a first round with six bands (16 runs, superseded and not counted in Table 3) behaved identically. Three properties of the capture make the problem hard for any deformation field: each instant is observed from a single viewpoint, frames are 0.13 s apart so flutter above 3.75 Hz is aliased, and the motion consists of thousands of small, independent rigid parts. These results characterise one representative design trained for 8,000 deformation iterations; dedicated dynamic methods or longer schedules may fare better, and SynthTree's per-leaf ground truth is what makes such claims testable. Whether the collapse is caused by the low frame rate or by the motion model is tested directly in Section 5.1.

Mitigation. The trimmed loss improves the posed F-score over plain static 3DGS in 16 of 16 tree–wind conditions (one-sided sign test $p = 1.5 \times 10^{-5}$, treating conditions as independent although they share four trees), raising precision and recall together, lowering the posed chamfer distance by 1.5 mm at 5 m/s and reducing floaters (Table 6); held-out PSNR is essentially unchanged. The gain, however, is largest in calm air (+0.046) and smaller at 5 m/s (+0.027), and the calm-to-5 m/s drop of the trimmed variant (0.284) is as large as that of plain 3DGS (0.265). Down-weighting poorly explained pixels is thus a cheap and consistently beneficial robustness measure, but it reduces the method/sensor component rather than the wind component of the error: once leaves move, the residual is not confined to a minority of outlier pixels that can be discarded.

Table 6: Static, trimmed-loss and deformable 3DGS (same implementation; noisy sensor; mean over 4 trees). F and CD against the rest pose and against the leaves posed at held-out frame times; PSNR on held-out views (deformed for the deformable model).

Wind	Method	F (rest)	F (posed)	CD rest [mm]	CD posed [mm]	Centroid posed [mm]	PSNR [dB]
calm	static 3DGS	0.463	0.456	7.1	7.1	5.1	31.3
	static 3DGS + trimmed loss	0.499	0.502	6.3	6.3	4.7	31.2
	deformable 3DGS	0.396	0.434	7.5	7.2	5.1	31.5
low	static 3DGS	0.301	0.308	9.7	9.8	7.9	30.5
	static 3DGS + trimmed loss	0.347	0.357	8.5	8.5	7.7	30.4
	deformable 3DGS	0.275	0.295	10.2	10.0	7.7	30.6
medium	static 3DGS	0.166	0.217	14.0	12.8	12.0	29.6
	static 3DGS + trimmed loss	0.192	0.258	12.6	11.2	11.8	29.5
	deformable 3DGS	0.158	0.216	14.3	12.8	12.1	29.7
high	static 3DGS	0.108	0.191	18.8	15.1	15.7	28.7
	static 3DGS + trimmed loss	0.126	0.218	17.6	13.6	15.9	28.5
	deformable 3DGS	0.098	0.187	19.2	15.3	15.8	28.8

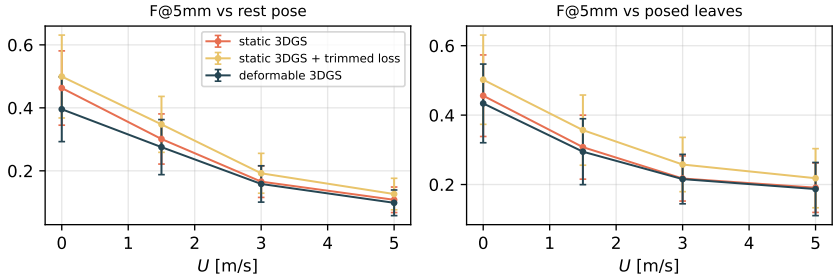


Figure 6: Leaf F-score versus wind speed against the rest pose (left) and against the posed leaves (right) for static, trimmed-loss and deformable 3DGS; mean ± std over 4 trees (one seed each); the spread reflects tree-to-tree variability.

5.1. Follow-up: sampling or modelling?

The frames used above are 0.13 s apart, and 83–87 % of the energy of the ground-truth leaf-normal trajectories lies above their 3.75 Hz Nyquist limit (closed-form wind model at 5 m/s, trees 1, 3, 5 and 7). The collapse of the deformable baseline could therefore be caused by temporal sampling rather than by the motion model. To separate the two, we re-rendered the four trees, calm and at 5 m/s, at the full 30 fps (600 frames per orbit; identical trees, wind phases and camera path) and trained every model on all frames (30 fps) and on every fourth frame (7.5 fps, the rate used above), so that both rates share the renders and the held-out views.

Per-part rigid motion. As a probe of motion structure, not as a new method, we add a deliberately simple per-part model in the spirit of superpoint and control-point Gaussians (Wan et al., 2024; Huang et al., 2024b). After the

static warm-up, every Gaussian is assigned to the nearest of 2,048 k-means cells of the warm-up geometry and to one of 32 coarser branch-scale cells. Each cell k follows its own rigid trajectory $x \mapsto R_k(t)\,(x - p_k) + p_k + v_k(t)$, whose rotation vector and translation are Fourier series in time up to 10 Hz, unlocked coarse to fine, with an acceleration penalty that is ten times stronger for branch cells; a Gaussian moves with its branch cell composed with its leaf-scale cell. Because cells own independent parameters, they cannot be smoothed into one global shift the way a deformation network can. An *oracle* variant replaces the k-means cells by the ground-truth leaf identities (one cell per leaf; Gaussians farther than 1 cm from every leaf keep k-means cells). It uses information no real capture provides and serves only as an upper bound for the grouping. All models use the schedule above (10,000 iterations, 2,000 static warm-up), noisy input and one seed (44 runs, Table 7). These runs use new renders and a newer release of the gsplat rasteriser (1.5.3 instead of 1.4.0); on tree 1 the deformable baseline at 7.5 fps reaches posed F@5mm of 0.449 (calm) and 0.137 (5 m/s) against 0.501 and 0.162 in Table 6, so absolute values are lower than above and only comparisons within Table 7 are drawn. [TODO: update after the seed and gsplat-version runs]

The collapse is a modelling failure. At 30 fps the deformable model still learns a uniform shift: its offsets have a median of 3.9 mm and a 95th percentile of 3.9 mm (95th-percentile-to-median ratio 1.0 on every tree), and its posed F-score at 5 m/s (0.160) stays below that of static 3DGS (0.172). Four times denser sampling does not prevent the collapse. The ground-truth motion indicates why a smooth field fails: the leaf-normal trajectories need 80–275 singular components to hold 90 % of their energy, and a single rigid transform per frame explains only 0.25–0.72 of the leaf-centroid motion. Independently fluttering leaves therefore cannot be represented by one smooth or low-rank deformation, which is also what shared motion bases assume (Wang et al., 2024).

Per-part motion removes the collapse but recovers little geometry. The per-cell model learns leaf-specific motion (offset median 2.6 mm, 95th percentile 10.8 mm; ratio 3.8–4.5). Compared with static 3DGS it reduces the wind part of the posed chamfer distance from 8.3 to 6.7 mm (−19 %) and raises held-out PSNR by 0.5 dB, but its posed F-score at 5 m/s (0.155) is not higher than that of static 3DGS (0.172; higher on 3 of 4 trees). On the needle tree, where one cell spans many needles, it is markedly worse (0.169 vs. 0.258), and it introduces spurious motion in calm air (calm F 0.112 vs. 0.233 there),

Table 7: 30 fps follow-up at 5 m/s: models trained on all frames (30 fps) or every fourth frame (7.5 fps) of the same renders; noisy sensor, one seed, mean over trees 1, 3, 5, 7. The wind part of CD is the calm-to-5 m/s increase of the posed chamfer distance. A uniform shift has equal median and 95th-percentile offsets. Absolute values are not comparable to Table 6 (new renders, newer rasteriser).

Method	fps	Trees	F (posed)	CD posed [mm]	Wind part of CD [mm]	Centroid posed [mm]	PSNR [dB]	Offset med / p95 [mm]
static 3DGS	30	4	0.172	15.8	8.3	16.1	28.76	–
deformable 3DGS	30	4	0.160	16.5	8.5	17.1	28.75	3.9 / 3.9
per-cell rigid motion	30	4	0.155	15.2	6.7	15.1	29.25	2.6 / 10.8
per-cell rigid motion	7.5	4	0.166	14.8	6.6	14.4	27.87	2.8 / 9.6
per-leaf rigid motion (oracle groups)	30	4	0.181	14.5	6.3	15.2	29.31	2.6 / 11.5

so part of the smaller wind term comes from a worse calm reconstruction rather than a better windy one. With oracle groups the posed F-score rises to 0.181 and the wind part of the chamfer distance falls by 23 %: even perfect per-leaf grouping leaves about three quarters of the wind-induced chamfer error.

Denser sampling does not help without blur. Trained on every fourth frame, the per-cell model reaches the same geometry (posed F 0.166, chamfer distance 14.8 mm) as with all frames (0.155, 15.2 mm). Without motion blur the frame rate is thus not what limits these reconstructions: evaluated at frame times, an aliased trajectory fits the samples as well as the true one, and each instant is still observed from a single viewpoint. The camera turns by about 2° during one 9 Hz flutter period, so the hand-held orbit is effectively monocular for leaf flutter (Gao et al., 2022). The remaining wind error is consistent with insufficient viewpoint coverage per instant; this is a hypothesis that capture with motion blur, higher frame rates or synchronised multi-view rigs can test, not a measured cause.

6. Discussion and limitations

All conclusions in this paper are drawn from synthetic data. The sensor-noise parameters and wind amplitudes are plausible defaults rather than values calibrated to a particular camera, plant and fan, so absolute error magnitudes may differ on real captures; the controlled comparisons (calm versus windy, clean versus noisy, method versus method), which are what the paired design measures, do not depend on these exact values. Confirming the error-versus-wind trends on real captures, using the calibration tooling of Section 3.5, is the most important next step. The generator also trades biomechanical fidelity for exactness: branch bending is piecewise rigid along skeleton segments, compound leaves move as single rigid units, leaves may interpenetrate, gusts are spatially uniform, and wind amplitudes and noise

parameters are defaults (Section 3.5). Motion blur and rolling shutter, which a real hand-held capture of moving leaves would add, were disabled in all experiments; the reported wind errors are therefore likely lower bounds for real RGB capture. When enabled, they affect RGB only, and ground truth refers to the frame-centre time, as for a global-shutter depth sensor. Rendered pixels are deterministic per machine but not across GPUs, whereas geometry, poses and ground truth are bit-identical. Our conclusions concern the implementations tested: splatfacto and depth-nerfacto from Nerfstudio and our gsplat-based static, trimmed-loss and deformable 3DGS. Surface-oriented splatting such as 2DGS (Huang et al., 2024a), other NeRF variants and RGB-D SLAM were not evaluated, and the benchmark covers four trees with one or two seeds, so per-tree effects are descriptive. Our deformable and per-part models are single, deliberately simple designs. The per-part model is a probe, not a contribution: closely related superpoint, control-point and motion-basis Gaussians exist (Wan et al., 2024; Huang et al., 2024b; Wang et al., 2024), and dedicated dynamic methods (Wu et al., 2024; Luiten et al., 2024) were not evaluated. The 30 fps follow-up (Section 5.1) uses one seed, new renders and a newer rasteriser release, so its absolute values are not directly comparable to the main benchmark. Capture with motion blur and higher frame rates, and synchronised multi-view rigs (which SynthTree supports through its static camera ring), are the natural next steps; the exported per-leaf transforms and optical flow allow their motion to be evaluated directly.

7. Conclusion

SynthTree provides paired calm/windy RGB-D sequences of plants with exact per-leaf ground truth and a protocol that separates occlusion, sensor and wind error. In simulation, under default sensor and wind models and across four leaf types and the methods tested, wind, not sensor noise, is the dominant source of geometric error for neural foliage reconstruction; image-space metrics conceal most of it, rest-pose evaluation over-states it for our static baseline, and neither the deformation-field baseline nor the robust photometric loss we tested recovers it. The collapse of the deformation field persists at 30 fps and is a failure of the motion model rather than of temporal sampling; per-part rigid motion removes it, but even with ground-truth leaf grouping it recovers only about a quarter of the wind-induced chamfer error. Confirming these trends on real captures is the next step.

Data and code availability

Code (generator, training and evaluation, MIT licence): `https://github.com/bibekgyawali2/synthtree`. The per-run results of all 108 runs are included in the repository as CSV files, and a single script (`scripts/paper/make_assets.py`) regenerates every table, figure and number in this paper from them; the Kaggle kernel scripts used for training are included under `kaggle/`. The 44 runs of the 30 fps follow-up are included as JSON files, and `scripts/paper/make_stage1_assets.py` regenerates its table and numbers; its renders (trees 1, 3, 4, 5, 7, calm and 5 m/s, 600 frames each) are regenerated by `configs/paper2_stage1.yaml`. Dataset (40 sequences, 9.2 GB, CC BY 4.0): `https://www.kaggle.com/datasets/bibekgyawali2/synthtree-v1`, available to editors and reviewers on request and to be made public and archived with a DOI upon publication; it can also be regenerated bit-for-bit (geometry, poses and ground truth) from the released configuration files.

CRediT authorship contribution statement

Bibek Gyawali: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare no competing interests.

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